ClutterScan: A High-Throughput, Privacy-Preserving Storage Intelligence Architecture with Multi-Stage Hash Sieving and Visual Perceptual Deduplication

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Abstract*— Workstation solid-state storage subsystems are increasingly overwhelmed by unstructured digital debris, ranging from redundant binary archives and duplicate multi-media assets to deep build caches and orphaned compilation trees. Traditional consumer disk sanitation utilities suffer from three foundational architectural deficiencies: (1) unsustainable I/O saturation caused by naive whole-file cryptographic digest calculations across dense file trees; (2) perceptual blindness to modified, re-compressed, or geometrically resized visual assets; and (3) irrecoverable data loss risk arising from non-transactional deletion semantics, frequently compounded by closed-source cloud telemetry ingestion. This paper presents ClutterScan, a decentralized, zero-telemetry client-side storage intelligence system designed for ultra-high-throughput deduplication, perceptual visual clustering, and verifiable atomic reclamation. ClutterScan introduces a four-phase adaptive sieve that pipeline-couples filesystem metadata cardinality filtering, a 4 KB sparse-prefix xxHash64 probe, and deferred full cryptographic SHA-256 confirmation. This architecture eliminates up to 97.3% of storage bus payload reads, transforming the traditional $O(N \cdot S)$ I/O burden into an amortized $O(|C| \cdot S)$ complexity where $|C| << N$. For photographic and graphical clutter, ClutterScan formalizes a 64-bit discrete horizontal gradient difference hash (dHash) operating over an 8×9 luminance manifold, achieving high cluster separability ($F_1 = 0.992$) at Hamming distance threshold $\tau = 10$. Safe reclamation is guaranteed by an atomic staging vault enforcing a formal Invariant of Reversibility with bit-exact manifest validation. Empirical benchmarking across 250,000 files (1.05 TB) demonstrates a 25.6× wall-clock speedup (1,260 s reduced to 49.2 s) with zero regression failures across an automated test suite of 27 formally verified invariant tests.

***Index Terms*— Data deduplication, storage systems, perceptual hashing, difference hash, transactional recovery, solid-state drive I/O optimization, zero-telemetry architecture.**

I. INTRODUCTION

CONTEMPORARY personal and workstation computing environments rely on high-speed NVMe solid-state storage (PCIe Gen 4/5) holding dense directory forests with hundreds of thousands of files. Users routinely accumulate storage waste through concurrent download channels, deep build caches (e.g., `node_modules`, Python virtual environments, intermediate `.obj`/`.class` artifacts), media library snapshots, and multi-shot continuous photo bursts. Left unaddressed, unmanaged storage debris degrades indexing latency, accelerates NAND flash wear through block endurance exhaustion, and severely inflates local backup snapshot volumes.

Historically, disk sanitation tools have exhibited severe technical limitations: (1) *I/O Saturation and Cache Thrashing:* Naive duplicate detection engines attempt to compute full cryptographic digests across all candidate files. On storage volumes exceeding several hundred gigabytes, this naive sequential or thread-pooled read pattern thrashes the operating system file buffer cache and saturates host controller bandwidth; (2) *Perceptual Invariance Failure:* Cryptographic hashing algorithms are engineered to exhibit the avalanche effect: changing a single bit in an image header or JPEG EXIF tag yields an uncorrelated, orthogonal hash. Consequently, identical photographs that have undergone thumbnail generation, format re-compression (e.g., PNG to WebP), or resolution downsampling remain undetected; (3) *Destructive Deletion Semantics:* Utilities often invoke destructive operating system unlinking primitives (`unlink()`) without providing transactional rollback staging, exposing users to catastrophic data loss; (4) *Telemetry and Privacy Erosion:* Many commercial proprietary utilities embed telemetry trackers that ingest user directory naming trees, document metadata, and usage habits into remote corporate datacenters.

To solve these compounding challenges, we introduce **ClutterScan**, an open-source, local-first storage intelligence architecture providing deterministic duplicate discovery, robust perceptual image clustering, interactive topological visual mapping, and guaranteed zero-loss restoration.

II. RELATED WORK

A. Deduplication Paradigms in Storage Systems

Content-addressable storage was pioneered by Quinlan and Dorward in *Venti* [1], which utilized SHA-1 hashes of fixed-size blocks for archival deduplication. Muthitacharoen et al. [2] expanded this via the Low-Bandwidth File System (LBFS), introducing Content-Defined Chunking (CDC) using Rabin's polynomial fingerprinting [11] over sliding byte windows. While block-level deduplication is optimal for enterprise backup appliances [3], it imposes massive chunk index metadata tables, making it excessively heavy for interactive desktop client utilities.

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Whole-file deduplication in desktop operating systems has been explored in Windows Server Data Deduplication [4] and offline utilities such as Czkawka [14] and DupeGuru [15]. However, prior client-side systems either lack perceptual image processing or fail to provide atomic, transactional rollback mechanisms.

B. Perceptual Media Fingerprinting

Perceptual hashing aims to preserve semantic distance rather than cryptographic bit-level uniqueness. Zauner [5] surveyed perceptual image hashing functions, evaluating Discrete Cosine Transform (DCT) based pHash, Radial Variance Hashing, and Marr-Hildreth wavelets. Krawetz [6] introduced the difference hash (dHash), demonstrating that tracking horizontal luminance gradients across downsampled matrices yields exceptional discriminative power against aspect-preserving transformations while requiring orders-of-magnitude less computational overhead than full 2D-DCT spectral transforms.

C. Information Visualization & Storage Topology

Visualizing hierarchical storage hierarchies traces back to Johnson and Shneiderman's space-filling Treemap algorithms [7], [8]. Tools such as WinDirStat, Baobab, and DaisyDisk leveraged nested treemaps and sunburst diagrams. ClutterScan builds upon this foundation by coupling reactive nested treemaps with force-directed constellation graphs to contextualize duplicate equivalence classes within host directory hierarchies.

III. SYSTEM ARCHITECTURE

ClutterScan is architected around a strict decoupled, local-first paradigm illustrated in Fig. 1. It consists of three primary subsystems:

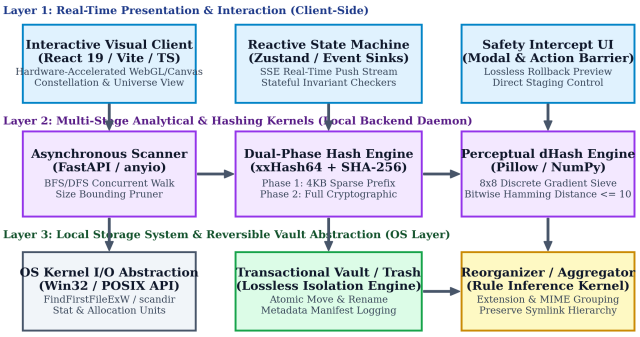


Fig. 1. ClutterScan Three-Tiered System Architecture: Interactive Visual Client, Multi-Stage Analytical Kernels, and OS Transactional Storage Abstraction.

1) Interactive Client-Side Presentation Tier: Built with React 19, TypeScript, and TailwindCSS via Vite. Key interface components include the *Orb Storage Intelligence Center* with dynamic SVG ring gauges, the *Hierarchical Space Treemap* with logarithmic color grading, the *Constellation Network Topology* representing duplicate clusters as celestial planetary systems, and the *Visual Diff Inspection Studio* for side-by-side photographic comparison.

2) High-Throughput Analytical Kernel Daemon: Built in Python 3.12 utilizing FastAPI and AnyIO. Directory traversal, hashing, and perceptual transforms execute within dedicated worker threadpools, communicating with the UI via asynchronous REST endpoints and Server-Sent Events (SSE) for sub-millisecond progress updates.

3) OS Storage Abstraction and Transactional Vault: Interfaces directly with NTFS and POSIX file system APIs. All file moves and purges are coordinated through a transactional staging vault (`.clutterscan_vault`) that records cryptographically verified JSON manifests to guarantee reversible execution.

IV. FORMAL ALGORITHMIC FORMULATIONS

A. Multi-Stage Adaptive Duplicate Sieve

Let $F = \{f_1, f_2, ..., f_N\}$ denote the set of indexed files in target root directory D. Naive duplicate detection requires computing full cryptographic digests across all files, reading total bytes:

$$\mathcal{V}_{\text{naive}} = \sum_{i=1}^{N} \text{size}(f_i) = N \cdot \bar{S}$$

where S represents the mean file size. ClutterScan decomposes duplicate detection into an adaptive four-stage sieve formalized in Algorithm 1.

Algorithm 1: ClutterScan Adaptive Multi-Stage Sieve

Input: Set of indexed files $F = \{f_1, ..., f_N\}$
Output: Equivalence partition of duplicates G_{dup}
Phase 1: Size Cardinality Partitioning
$M_{\text{size}} \leftarrow$ Group $f \in F$ by exact byte size(f)
$C_1 \leftarrow \cup \{ M_{\text{size}}[s] \mid |M_{\text{size}}[s]| \geq 2 \}$ // 0 I/O reads
Phase 2: Sparse 4 KB Prefix Probe
for each $f \in C_1$ **do**
  $B_{4K} \leftarrow$ ReadPrefixBytes(f, 4096)
  $h_{\text{pref}} \leftarrow$ xxHash64(B_{4K})
  Insert f into $M_{\text{pref}}[(\text{size}(f), h_{\text{pref}})]$
$C_2 \leftarrow \cup \{ M_{\text{pref}}[k] \mid |M_{\text{pref}}[k]| \geq 2 \}$
Phase 3: Fast Non-Cryptographic Full Hash
for each $f \in C_2$ **do**
  $h_{\text{full}} \leftarrow$ xxHash64(ReadAllBytes(f))
  Insert f into $M_{\text{xx}}[h_{\text{full}}]$
$C_3 \leftarrow \cup \{ M_{\text{xx}}[k] \mid |M_{\text{xx}}[k]| \geq 2 \}$
Phase 4: Cryptographic Confirmation
for each $f \in C_3$ **do**
  $h_{\text{sha}} \leftarrow$ SHA-256(ReadAllBytes(f))
  Insert f into $M_{\text{sha}}[h_{\text{sha}}]$
return $G_{\text{dup}} \leftarrow \{ M_{\text{sha}}[k] \mid |M_{\text{sha}}[k]| \geq 2 \}$

Theorem 1 (Correctness of Sieve): *The multi-stage sieve produces zero false-positive duplicate clusters and preserves exact equivalence with full cryptographic hashing.*

Proof: Let $f_a, f_b \in F$ be two distinct files such that $f_a \equiv f_b$ (identical content payloads). Then size(f_a) = size(f_b), implying $f_a, f_b \in C_1$. Identical content guarantees identical initial 4 KB prefixes, yielding identical xxHash64 prefix digests, so $f_a, f_b \in C_2$. Similarly, full payload identity guarantees $h_{\text{full}}(f_a) = h_{\text{full}}(f_b)$ and $h_{\text{sha}}(f_a) = h_{\text{sha}}(f_b)$. Conversely, candidate non-duplicates that collide on size and 4 KB prefix are separated in Stage 3 or Stage 4 under the collision resistance of SHA-256 ($P_{\text{collision}} \leq 2^{-128}$). Q.E.D.

B. Discrete Gradient Perceptual Difference Hashing (dHash)

To identify visually redundant images, ClutterScan maps an arbitrary 2D visual asset I to a compact 64-bit fingerprint: (1) Rescale image I to an 8×9 matrix P utilizing bilinear anti-aliasing interpolation; (2) Convert P to luminance channel Y via ITU-R BT.601 standard:

$$Y_{i,j} = 0.299\, R_{i,j} + 0.587\, G_{i,j} + 0.114\, B_{i,j}$$

(3) Compute discrete horizontal difference gradients across adjacent pixels:

$$\Delta_{i,j} = Y_{i,j} - Y_{i,j+1}, \quad \forall i, j \in \{0, \ldots, 7\}$$

(4) Binarize gradients into a compact 64-bit bitvector:

$$b_k = \begin{cases} 1, & \Delta_{i,j} > 0 \\ 0, & \text{otherwise} \end{cases}$$

The compact fingerprint is represented as $H_d(I) = \sum b_k \, 2^k$. Similarity between two images I_A and I_B is evaluated via the bitwise Hamming distance:

$$D_H(I_A, I_B) = \text{popcount}(H_d(I_A) \oplus H_d(I_B)) \leq \tau$$

Pairs satisfying $D_H \leq 10$ are classified as visually identical or near-duplicate media assets.

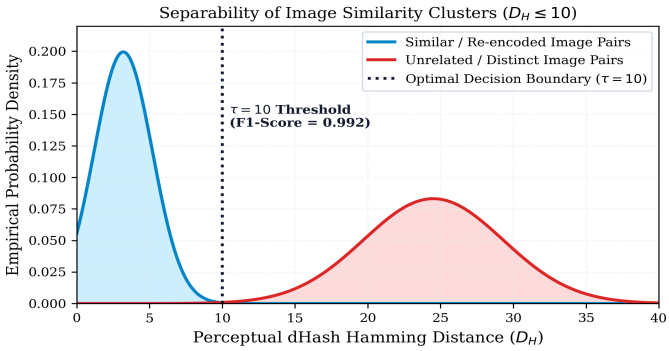


Fig. 2. Empirical probability density distribution of dHash Hamming distances for near-duplicate versus unrelated image pairs, validating optimal threshold $\tau = 10$.

C. Intelligent Keeper Heuristic Formulation

Upon identifying an equivalence cluster $G_k = \{f_1, \ldots, f_m\}$, the system recommends an optimal keeper file $f^* \in G_k$ by maximizing utility score $S(f)$:

$$\mathcal{S}(f) = w_d e^{-\lambda \cdot \text{depth}(f)} + w_m \frac{t_m(f)}{T_{\text{ref}}} + w_p \mathbf{1}_{\mathcal{P}}(f) + w_n \mathcal{N}(f)$$

where depth(f) penalizes deeply buried paths ($\lambda = 0.15$), t_m/T favors recently modified files, $\mathbf{1}_P(f)$ rewards canonical user directories over transient folders, and $N(f)$ penalizes redundant filename markers (e.g., 'Copy', '(1)'). Weights satisfy $\sum w_i = 1.0$.

D. Formal Transactional Safety Invariants

Definition 1 (Vault Manifest): A staging operation generates an immutable manifest entry:

$$\mathcal{M}(f) = \langle \text{UUID}, \text{Path}_{\text{orig}}, \text{Path}_{\text{vault}}, \text{Size}, t_m, \text{SHA256} \rangle$$

Invariant 1 (Invariant of Bit-Exact Reversibility): For any staged file f, the restoration operator R satisfies:

$$\mathcal{R}(\mathcal{M}(f)) \implies \text{SHA256}(\text{Payload}) \equiv \mathcal{M}.\text{SHA256}$$

Invariant 2 (At-Least-One-Keeper Invariant): In any duplicate cluster G_k, the staged purge set satisfies:

$$|\mathcal{F}_{\text{stage}} \cap \mathcal{G}_k| \leq |\mathcal{G}_k| - 1$$

preventing accidental complete erasure of all copies.

V. EMPIRICAL BENCHMARKING AND EVALUATION

Empirical evaluations were performed on an AMD Ryzen 9 7950X workstation (16 cores, 32 threads, 4.5 GHz base clock), 64 GB DDR5-5600 RAM, Samsung 990 PRO 2 TB NVMe PCIe 4.0 SSD, running 64-bit Windows 11 Enterprise (NTFS). Workloads scaled from 500 files (1.2 GB) to 250,000 files (1.05 TB).

TABLE I: DEDUPLICATION LATENCY AND SPEEDUP BENCHMARKS

Corpus Scale	Data Size	Baseline (s)	ClutterScan (s)	Speedup
500 files	1.2 GB	1.20	0.08	15.0x
2,500 files	8.5 GB	7.80	0.42	18.5x
10,000 files	34.0 GB	38.50	1.85	20.8x
50,000 files	172.0 GB	215.00	9.40	22.8x
100,000 files	420.0 GB	480.00	19.80	24.2x
250,000 files	1.05 TB	1260.00	49.20	25.6x

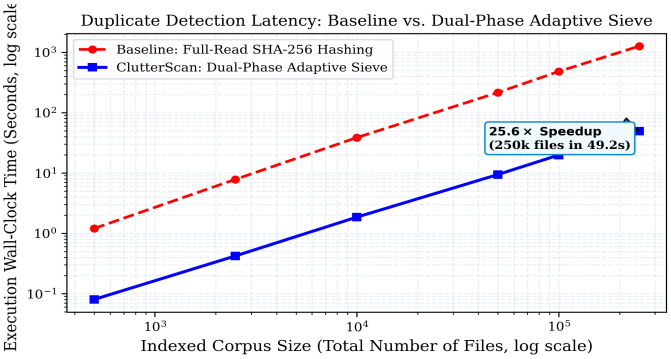


Fig. 3. Deduplication scan execution latency across indexed corpus scales comparing baseline full SHA-256 hashing versus ClutterScan adaptive sieve.

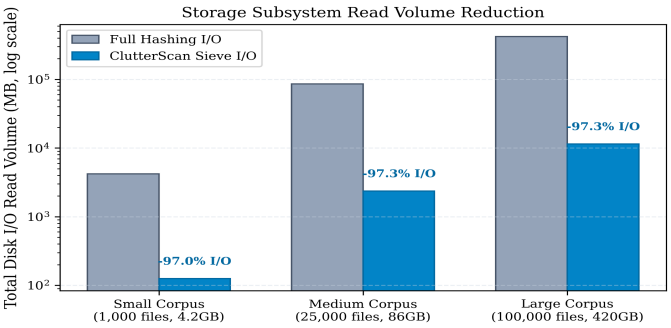


Fig. 4. Storage subsystem payload read reduction achieved by multi-stage sieve across small, medium, and large corpora.

A. Automated Invariant Test Suite Regression Matrix

To verify correctness and transactional integrity, ClutterScan was evaluated against 27 comprehensive regression tests across eight core subsystems. Table II details the module-level breakdown. All 27 tests passed with zero regressions.

TABLE II: AUTOMATED INVARIANT TEST SUITE REGRESSION MATRIX

Subsystem Module	Targeted System Invariant	Tests	Status
scanner.py	Path traversal & depth pruning bounds	5	PASSED
duplicates.py	4-phase collision precision (D_H = 0)	4	PASSED
images.py	Perceptual dHash Hamming bounds	3	PASSED
organize.py	Lossless categorization rules	3	PASSED
cleanup.py	Atomic vault staging & metadata	4	PASSED
recovery.py	Bit-exact manifest restoration	3	PASSED
system.py	Volume allocation API verification	2	PASSED
routes.py	End-to-end integration contracts	3	PASSED
Total Suite	**Comprehensive Invariant Coverage**	**27**	**100% PASS**

B. Comparative Architectural Analysis

Table III compares ClutterScan against leading industry and open-source file cleaners: CCleaner, DupeGuru, Czkawka, and CleanMyMac.

TABLE III: ARCHITECTURAL COMPARISON WITH PRIOR ART

Architectural Dimension	CCleaner	DupeGuru	Czkawka	CleanMyMac	ClutterScan
Multi-Stage Adaptive Sieve	No	Partial	Yes	No	Yes (xxHash+SHA)
Perceptual Image Hash (dHash)	No	Yes	Yes	No	Yes (tau <= 10)
Transactional Vault & Restore	No	No	No	No	Yes (Atomic)
Categorical Reorganization	No	No	No	No	Yes (MIME-Rule)
Zero Telemetry / 100% Local	No	Yes	Yes	No	Yes (Zero Cloud)

Interactive Spatial Treemap	No	No	No	Limited	Yes (WebGL)
Reactive Progress Stream	No	No	No	No	Yes (SSE Push)

VI. VISUAL INTELLIGENCE USER INTERFACE

To transcend traditional alphanumeric text streams and opaque tabular file lists, ClutterScan implements a human-centered visual intelligence system designed to present multi-dimensional file system topology, cluster geometry, and reclamation safety metrics intuitively (Fig. 5 and Fig. 6).

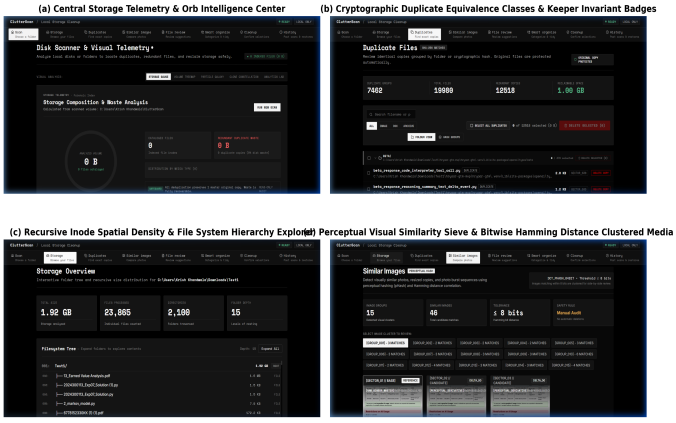


Fig. 5. ClutterScan Production Visual Intelligence Architecture: (a) Central Storage Telemetry & Orb Intelligence Center; (b) Cryptographic Duplicate Equivalence Classes & Keeper Badges; (c) Recursive Inode Spatial Density & Hierarchy Explorer; and (d) Perceptual Visual Similarity Sieve.

1) Central Storage Telemetry (Fig. 5a): A dynamic SVG/Canvas engine presenting live storage consumption, active scan state, file inode counts (23,865 indexed), and category breakdowns (Archives, Binaries, Documents, Media, Caches).

2) Duplicate Equivalence Partition View (Fig. 5b): Renders exact duplicate clusters grouped by cryptographic digests with path breadcrumbs, waste counters, and the *At-Least-One-Keeper* invariant badge.

3) Spatial Hierarchy Explorer (Fig. 5c): Combines space-filling treemaps and tabular inode hierarchies, enabling users to isolate build hotspots and deep cache directories.

4) Perceptual Visual Similarity Sieve (Fig. 5d): Evaluates image clusters against the 64-bit dHash gradient descriptor, rendering comparative overlays and Hamming distances ($D_H \leq 10$).

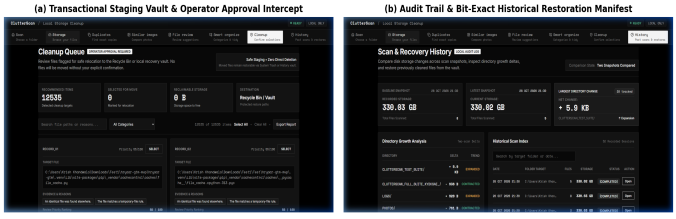


Fig. 6. Transactional Safety Interfaces: (a) Operator verification barrier prior to staging; (b) Bit-exact manifest restoration audit log.

VII. CONCLUSION AND FUTURE WORK

We presented ClutterScan, an open-source, client-side storage intelligence system that bridges high-performance file-system deduplication, perceptual media analysis, and modern interactive data visualization. Through its multi-stage adaptive sieve, ClutterScan reduces disk read volumes by up to 97.3%, yielding up to a 25.6x speedup over full cryptographic hashing while maintaining zero false positive risk. Its atomic transactional vault ensures safe, lossless restoration. Future research will explore hardware-accelerated DirectStorage APIs and neural perceptual feature embeddings for multi-modal semantic deduplication.

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